Computer Networks

500+ One-Liners for Rapid Revision

IBPS SO (IT Officer) Examination - Complete Preparation Series

100% Syllabus Coverage - MCQ-Oriented - Zero Filler

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Network Fundamentals

- A network needs **2** autonomous devices minimum connected to **share resources** and **exchange data**.
- There are **4** main network goals: **resource sharing**, **reliability**, **scalability**, and **cost saving**.
- A **distributed system** differs from a network by **1** thing: an extra **software** abstraction layer hiding the network.
- There are **4** network types by size: **PAN**, **LAN**, **MAN**, and **WAN**.
- The smallest network type is **PAN**, spanning about **1-10 m**, exemplified by **Bluetooth**.
- A **LAN** spans up to **1 km** and uses **Ethernet** or **Wi-Fi** technology.
- A **MAN** spans a city about **10-100 km**, exemplified by **cable TV** and **WiMAX**.
- A **WAN** spans **countries/continents** and uses **public carrier** links like the **Internet**.
- The increasing size order of networks is **PAN < LAN < MAN < WAN**.
- There are **6** common topologies: **bus**, **star**, **ring**, **mesh**, **tree**, and **hybrid**.
- A full mesh of **n** nodes needs **n(n-1)/2** links and **n-1** ports per node.
- A mesh of **5** nodes requires **10** links.
- A **bus** topology needs **1** backbone cable and **2** terminators at the ends.
- A **star** topology uses **1** central hub/switch and **n** cables for n nodes.
- A **star** topology's single point of failure is the **central hub**.
- A **ring** topology connects each node to **2** neighbors and uses **token passing**.
- The **mesh** topology is the most **reliable** but the most **expensive**.
- There are **3** transmission modes: **simplex**, **half-duplex**, and **full-duplex**.
- **Simplex** allows communication in **1** direction only, like a **keyboard**.
- **Half-duplex** allows **both** directions but **one at a time**, like a **walkie-talkie**.
- **Full-duplex** allows **both** directions **simultaneously**, like a **telephone**.
- There are **2** service types: **connection-oriented** (TCP) and **connectionless** (UDP).
- A connection-oriented service has **3** phases: **setup**, **transfer**, and **teardown**.
- **Transmission delay** equals **L/B**, depending on **packet size** and **bandwidth**.
- **Propagation delay** equals **d/v**, depending on **distance** and **medium speed**.
- Propagation speed in copper/fiber is about **2x10^8 m/s** versus **3x10^8** in vacuum.
- There are **4** delay components: **transmission**, **propagation**, **queuing**, and **processing**.
- **Queuing delay** depends on **congestion** and is the most **variable**.
- **Bandwidth-delay product** equals **bandwidth x RTT** and gives max **bits in flight**.
- **RTT** equals **2 x propagation delay** for a round trip.
- **Throughput** is always **<= bandwidth** and measures actual successful delivery.
- Doubling bandwidth halves **transmission delay** but not **propagation delay**.

Layered Architecture

- Layering provides **4** benefits: **modularity**, **abstraction**, **interoperability**, and easy **maintenance**.
- A **service** is what a layer offers **above**; a **protocol** governs **peer** communication.
- An **interface** defines how to **access** a layer's service between **adjacent** layers.
- The **OSI** model has **7** layers, defined by **ISO** in **1984**.
- OSI layers bottom-up are **Physical**, **Data Link**, **Network**, **Transport**, **Session**, **Presentation**, **Application**.
- The OSI mnemonic bottom-up is "**Please Do Not Throw Sausage Pizza Away**".
- The **Physical** layer (L1) transmits **raw bits** and its PDU is the **bit**.
- The **Data Link** layer (L2) does **framing** and uses **MAC** addresses; PDU = **frame**.
- The **Network** layer (L3) does **routing** with **IP** addresses; PDU = **packet**.

- The **Transport** layer (L4) gives **process-to-process** delivery; PDU = **segment**.
- The **Session** layer (L5) handles **dialog control** and **synchronization**.
- The **Presentation** layer (L6) handles **translation**, **encryption**, and **compression**.
- The **Application** layer (L7) provides **user services** like **HTTP** and **SMTP**.
- The **TCP/IP** model has **4** layers: **Application**, **Transport**, **Internet**, and **Link**.
- TCP/IP's **Application** layer combines OSI layers **5**, **6**, and **7**.
- The **Internet** layer maps to OSI **Network** layer and runs **IP**.
- OSI is **prescriptive** while TCP/IP is **descriptive**.
- Only the **Data Link** layer adds both a **header** and a **trailer**.
- The **4** PDU names down the stack are **segment**, **packet**, **frame**, and **bits**.

Physical Layer

- There are **2** signal types: **analog** (continuous) and **digital** (discrete).
- There are **2** media categories: **guided** (wired) and **unguided** (wireless).
- The **3** guided media are **twisted pair**, **coaxial**, and **fiber optic**.
- **Twisted pair** reduces **crosstalk** by twisting and comes in **UTP** and **STP**.
- **Fiber optic** has the **highest** bandwidth and immunity to **EMI**.
- Fiber has **2** modes: **single-mode** (laser, long) and **multi-mode** (LED, short).
- **Nyquist** bit rate for a noiseless channel is **$C = 2B \log_2(L)$**.
- **Shannon** capacity for a noisy channel is **$C = B \log_2(1 + S/N)$**.
- **SNR in dB** equals **$10 \log_{10}(S/N)$**.
- **Bit rate** equals **baud rate x $\log_2(L)$** and baud is always **$\le$ bit rate**.
- **Manchester** encoding is **self-clocking** but needs **2x** the bandwidth.
- **NRZ** is bandwidth-efficient but loses **synchronization** on long runs.
- There are **3** multiplexing types: **FDM** (frequency), **TDM** (time), and **WDM** (wavelength).
- **Statistical TDM** allocates slots **on demand** with no wasted slots.
- **WDM** is essentially **FDM** applied to **fiber optic** light wavelengths.

Switching

- There are **3** switching techniques: **circuit**, **packet**, and **message** switching.
- **Circuit switching** has **3** phases: **setup**, **data transfer**, and **teardown**.
- **Circuit switching** reserves bandwidth even when **idle**, used in **PSTN**.
- **Packet switching** has **2** types: **datagram** and **virtual circuit**.
- In **datagram** switching each packet carries the **full destination address**.
- **Virtual circuits** carry a small **VC label** and come in **PVC** and **SVC** types.
- **Message switching** uses **store-and-forward** of the whole message and is now **obsolete**.
- **Datagram** packets may arrive **out of order**, while VC preserves **order**.

Data Link Layer

- The **Data Link** layer provides **node-to-node** delivery and has **5** functions: **framing**, **addressing**, **flow control**, **error control**, and **access control**.
- IEEE 802 splits the DLL into **2** sublayers: **LLC** (upper) and **MAC** (lower).
- The **LLC** sublayer is defined by **IEEE 802.2** and interfaces to the **network** layer.

- The **MAC** sublayer controls **medium access** and holds **MAC** addressing.
- There are **2** framing approaches: **fixed-size** (ATM cells) and **variable-size**.
- **ATM cells** are a fixed **53** bytes, needing no delimiters.
- **Byte count** framing is the least **robust** because a corrupted count desynchronizes.
- **Byte stuffing** inserts an **ESC** byte before any data byte matching the **FLAG**.
- **Bit stuffing** inserts a **0** after **5** consecutive **1**s in the data.
- The HDLC flag is **01111110** which contains **6** consecutive ones.

Error Detection and Correction

- There are **2** error types: **single-bit** and **burst** errors.
- A **burst** error involves **2 or more** corrupted bits.
- **Hamming distance** is computed by **XOR** of two codewords then counting **1**s.
- To detect **s** errors a code needs minimum Hamming distance **s+1**.
- To correct **t** errors a code needs minimum Hamming distance **2t+1**.
- A code with **d_min = 3** can detect **2** errors and correct **1** error.
- **Simple parity** detects only an **odd** number of bit errors and has **d_min = 2**.
- **Simple parity** cannot detect **2-bit** (even) errors nor correct anything.
- **2D parity** can **correct** a single-bit error at the **row-column** intersection.
- The **Internet checksum** uses **16-bit** words and **1's complement** arithmetic.
- A correct checksum produces a receiver sum of **all 1s** (zero in 1's complement).
- **CRC** uses **modulo-2** (XOR) division by a **generator polynomial**.
- A generator of degree **r** produces **r** CRC/FCS bits.
- **CRC** detects all burst errors of length **<= r**.
- **CRC** detects odd errors if the generator contains the factor **(x+1)**.
- **CRC-32** used in **Ethernet** has **r = 32** bits.
- **Hamming code** places parity bits at positions that are **powers of 2**: **1, 2, 4, 8**.
- The Hamming redundant-bit rule is **2^r >= m + r + 1**.
- For **7** data bits, Hamming needs **4** parity bits, forming an **(11,7)** code.
- The Hamming **syndrome** directly gives the **position** of the erroneous bit.

Flow and Error Control Protocols

- **Stop-and-Wait** has a window of **1** and efficiency **1/(1+2a)**.
- The parameter **a** equals **Tp/Tt** (propagation over transmission delay).
- **Stop-and-Wait ARQ** needs only **1** sequence-number bit (**0/1**).
- **Go-Back-N** sender window is **2^m - 1** and receiver window is **1**.
- **Selective Repeat** window is **2^(m-1)** for both sender and receiver.
- **Go-Back-N** retransmits **all** frames after a loss; **Selective Repeat** retransmits **only** the lost one.
- For **m=3** bits, GBN window = **7** and SR window = **4**.
- Sliding window efficiency is **N/(1+2a)**, capped at **1**.
- The optimal window size equals **1 + 2a** to keep the pipe full.
- **Piggybacking** attaches an **ACK** onto an outgoing **data** frame in bidirectional links.
- **Go-Back-N** uses **cumulative** ACKs; **Selective Repeat** uses **selective** ACKs.
- **Selective Repeat** requires the receiver to **buffer** out-of-order frames.

Medium Access Control

- There are **2** allocation types: **static** (FDM/TDM) and **dynamic**.
- **Pure ALOHA** has a vulnerable period of **2T** and max efficiency **18.4%**.
- **Slotted ALOHA** has a vulnerable period of **1T** and max efficiency **36.8%**.
- **Pure ALOHA** throughput is **S = G e^(-2G)**, peaking at **G=0.5**.
- **Slotted ALOHA** throughput is **S = G e^(-G)**, peaking at **G=1**.
- **CSMA** means **carrier sense multiple access**: listen before transmit.
- There are **3** CSMA persistence methods: **1-persistent**, **non-persistent**, and **p-persistent**.
- **CSMA/CD** is used in **wired Ethernet** and detects collisions while transmitting.
- **CSMA/CA** is used in **Wi-Fi (802.11)** and avoids collisions.
- **CSMA/CD** needs minimum frame size satisfying **Tt >= 2Tp**.
- After **k** collisions, backoff is chosen from **0 to 2^k - 1**, capped at **k=10**.
- Ethernet allows a maximum of **16** transmission attempts before giving up.
- There are **3** controlled-access methods: **reservation**, **polling**, and **token passing**.
- There are **3** channelization methods: **FDMA**, **TDMA**, and **CDMA**.
- **CDMA** separates users by orthogonal **codes** on the same frequency and time.
- Slotted ALOHA **doubles** pure ALOHA efficiency from **18.4%** to **36.8%**.

Ethernet and LAN Devices

- The **IEEE 802.3** Ethernet frame has **7** fields including **preamble** and **FCS**.
- The Ethernet **preamble** is **7** bytes and the **SFD** is **1** byte.
- The minimum Ethernet frame is **64** bytes to support **CSMA/CD**.
- The maximum Ethernet frame is **1518** bytes with payload (MTU) **1500**.
- The minimum Ethernet payload is **46** bytes, padded if smaller.
- The Ethernet **FCS** field uses **CRC-32** and is **4** bytes.
- A **MAC address** is **48** bits, shown as **12** hex digits.
- The first **24** bits of a MAC address are the **OUI** (vendor ID).
- A **broadcast** MAC address is **FF:FF:FF:FF:FF:FF** (all ones).
- A **multicast** MAC has the **LSB** of the first byte set to **1**.
- Ethernet speeds are **10 Mbps**, Fast **100 Mbps**, Gigabit **1 Gbps**, and **10 Gbps**.
- A **hub** operates at layer **1** with **1** collision domain and **1** broadcast domain.
- A **switch** operates at layer **2** giving each port its own **collision** domain.
- A **router** operates at layer **3** and separates **broadcast** domains.
- A **gateway** operates up to layer **7** as a **protocol converter**.
- A **switch** learns from **source** MAC and forwards by **destination** MAC.
- Only a **router** or **VLAN** can break a **broadcast** domain.
- **STP (802.1D)** prevents layer-**2** loops by electing a **root bridge**.
- **VLAN** tagging uses **802.1Q** with a **4**-byte tag and **12**-bit VLAN ID.
- **IEEE 802.11** is **Wi-Fi** and uses **CSMA/CA**.

Network Layer

- The **Network** layer provides **3** functions: **logical addressing**, **routing**, and **forwarding**.
- **Routing** builds the table (control plane); **forwarding** uses it (data plane).

- The Internet's network layer is **connectionless** and **best-effort** via **IP**.

IPv4 Addressing

- An **IPv4** address is **32** bits = **4** octets in **dotted decimal**.
- IPv4 has about **4.3 billion** (**2^32**) total addresses.
- There are **5** IPv4 classes: **A**, **B**, **C**, **D**, and **E**.
- Class **A** has default mask **/8** and first octet range **1-126**.
- Class **B** has default mask **/16** and first octet range **128-191**.
- Class **C** has default mask **/24** and first octet range **192-223**.
- Class **D** (**224-239**) is for **multicast**; class **E** (**240-255**) is **reserved**.
- The **loopback** address is **127.0.0.1** in the **127.0.0.0/8** block.
- A **network** address has host bits all **0**; a **broadcast** has host bits all **1**.
- The limited broadcast address is **255.255.255.255**.
- There are **3** private ranges: **10.0.0.0/8**, **172.16.0.0/12**, and **192.168.0.0/16**.
- **APIPA** link-local addresses are in **169.254.0.0/16**.
- **CIDR** uses a **/n** prefix; block size equals **2^(32-n)** addresses.
- Usable hosts per subnet equal **2^h - 2** (minus network and broadcast).
- Number of subnets from borrowing **s** bits equals **2^s**.
- The subnet block size (increment) equals **256 - mask octet value**.
- A **/30** subnet has **4** addresses and **2** usable hosts (point-to-point).
- A **/24** subnet has **256** addresses and **254** usable hosts.
- A **/31** (RFC 3021) allows **2** hosts with **no** broadcast.
- **VLSM** allows subnets of **different** sizes, allocated **largest first**.
- **Supernetting** aggregates contiguous networks with a **shorter** prefix.
- Network address equals **IP AND mask**; broadcast equals **network OR (NOT mask)**.

IPv4 Datagram

- The **IPv4** header is **20** bytes minimum and **60** bytes maximum.
- **IHL** counts header length in **4-byte** words, min **5** and max **15**.
- The **Total Length** field is **16** bits, allowing up to **65535** bytes.
- **TTL** is decremented by each **router** and at **0** triggers ICMP **Time Exceeded**.
- The **Protocol** field values are **ICMP=1**, **TCP=6**, **UDP=17**, and **IGMP=2**.
- The IPv4 **header checksum** covers only the **header** and is recomputed each **hop**.
- **Ethernet MTU** is **1500** bytes, triggering fragmentation of larger packets.
- The **3** IPv4 flags are reserved, **DF** (don't fragment), and **MF** (more fragments).
- **MF=0** marks the **last** fragment of a datagram.
- The **fragment offset** is measured in **8**-byte units.
- Each non-last fragment's data must be a multiple of **8** bytes.
- IPv4 reassembly happens only at the **destination**.

IPv6

- An **IPv6** address is **128** bits in **8** groups of **4** hex digits.
- The IPv6 header is a fixed **40** bytes with **no** checksum.

- IPv6 routers do **not** fragment; only the **source** fragments.
- IPv6 renames **TTL** to **Hop Limit** and **Protocol** to **Next Header**.
- The **::** shorthand for zeros can appear only **once** in an IPv6 address.
- There are **3** IPv6 transition methods: **dual stack**, **tunneling**, and **translation**.

Routing Algorithms

- There are **2** routing categories: **static** and **dynamic**.
- Routing metrics include **hop count**, **bandwidth**, **delay**, **cost**, and **reliability**.
- **Dijkstra's algorithm** finds shortest paths in **O(V^2)** or **O((V+E)log V)** with a heap.
- **Dijkstra** requires **non-negative** edge weights.
- **Flooding** forwards a packet on **all** links except the **incoming** one.
- **Distance Vector** routing is based on **Bellman-Ford** and shares the **whole table** with **neighbors**.
- **Distance Vector** suffers from the **count-to-infinity** problem.
- **Split horizon** and **poison reverse** mitigate **count-to-infinity**.
- **RIP** uses **hop count** with max **15** and infinity **16**, on **UDP port 520**.
- RIP sends updates every **30** seconds.
- **Link State** routing floods **LSPs** to **all** routers and runs **Dijkstra** locally.
- **OSPF** is a link-state protocol with **cost** metric and IP protocol number **89**.
- **BGP** is a **path-vector** protocol using **TCP port 179**, the Internet's routing protocol.
- **IGP** runs within an **AS** (RIP/OSPF); **EGP** runs between AS (**BGP**).
- **Link State** converges **faster** than **Distance Vector**.

IP Support Protocols

- **ARP** resolves a known **IP** to a **MAC** using a **broadcast** request and **unicast** reply.
- **ARP** results are stored in the **ARP cache** with a timeout.
- **RARP** maps **MAC** to **IP** and is replaced by **DHCP**.
- **ICMP** is for **error reporting**, has IP protocol number **1**, and includes **ping**.
- **ping** uses ICMP **echo request/reply** to test reachability.
- **traceroute** discovers hops by incrementing **TTL** and reading **Time Exceeded**.
- **DHCP** uses **UDP port 67** (server) and **68** (client).
- The DHCP process **DORA** stands for **Discover**, **Offer**, **Request**, and **Acknowledge**.
- DHCP lease renewal occurs at **T1=50%** and **T2=87.5%** of the lease.
- There are **3** NAT types: **static**, **dynamic**, and **PAT/overloading**.
- **PAT** maps many private addresses to **one** public address using **port** numbers.

Transport Layer

- The **Transport** layer gives **process-to-process** delivery, unlike the network layer's **host-to-host**.
- A **port number** is **16** bits, ranging **0-65535**.
- A **socket** is defined as **IP address + port number**.
- A TCP connection is identified by a **4-tuple**: **source IP**, **source port**, **dest IP**, **dest port**.
- **Multiplexing** gathers data from many sockets; **demultiplexing** delivers to the right socket.
- There are **3** port ranges: **well-known 0-1023**, **registered 1024-49151**, **ephemeral 49152-65535**.

UDP

- The **UDP** header is a fixed **8** bytes with **4** fields.
- The **4** UDP header fields are **source port**, **destination port**, **length**, and **checksum**.
- The UDP **checksum** is optional in **IPv4** but mandatory in **IPv6**.
- **UDP** is used by **DNS**, **DHCP**, **TFTP**, **VoIP**, and **streaming**.
- **UDP** has **no** sequencing, flow control, or congestion control.

TCP

- The **TCP** header is **20** bytes minimum and **60** bytes maximum.
- TCP has **6** main control flags: **URG**, **ACK**, **PSH**, **RST**, **SYN**, and **FIN**.
- TCP is **byte-oriented**; the sequence number is the **first byte** number in a segment.
- The TCP **ACK number** is the **next expected byte** and is **cumulative**.
- TCP connection setup uses a **3-way** handshake: **SYN**, **SYN-ACK**, **ACK**.
- TCP connection teardown is **4-way**: **FIN**, **ACK**, **FIN**, **ACK**.
- The **TIME-WAIT** state lasts **2*MSL** before closing.
- TCP **flow control** protects the **receiver** via the **advertised window (rwnd)**.
- The TCP sender is limited by **min(rwnd, cwnd)**.
- TCP **congestion control** protects the **network** using the **congestion window (cwnd)**.
- **Slow start** begins cwnd at **1 MSS** and **doubles** each RTT until **ssthresh**.
- **Congestion avoidance (AIMD)** grows cwnd **linearly** (+1 MSS per RTT).
- On **timeout**, ssthresh becomes **cwnd/2** and cwnd resets to **1**.
- **Fast retransmit** triggers on **3 duplicate ACKs** without waiting for timeout.
- **Fast recovery** (TCP Reno) sets cwnd to **ssthresh** after 3 dup ACKs.
- **Slow start** is misleadingly named because it is actually **exponential**.
- TCP is **connection-oriented** and **reliable**; UDP is **connectionless** and **unreliable**.

DNS

- **DNS** maps **domain names** to **IP addresses** in a **hierarchical** database.
- The DNS hierarchy has **3** levels: **root**, **TLD**, and **authoritative**.
- There are **13** logical root server clusters worldwide.
- DNS record **A** maps to **IPv4** and **AAAA** maps to **IPv6**.
- DNS record **MX** identifies the **mail server** and **NS** the **name server**.
- DNS record **CNAME** is an **alias** and **PTR** is a **reverse** lookup.
- DNS uses **UDP port 53** for queries and **TCP port 53** for **zone transfers**.
- There are **2** DNS resolution modes: **recursive** and **iterative**.
- **Recursive** resolution returns the **final** answer; **iterative** returns a **referral**.

Email Protocols

- Email architecture has **3** agents: **MUA**, **MTA**, and **MDA**.
- **SMTP** is a **push** protocol on **TCP port 25**.
- **POP3** pulls and usually **deletes** mail from the server on **TCP port 110**.
- **IMAP** pulls and **syncs** mail across devices on **TCP port 143**.

- **MIME** extends email to carry **non-ASCII** data and **attachments**.
- The secure ports are **POP3S 995**, **IMAPS 993**, and **SMTPS 465**.

Web (HTTP/HTTPS)

- **HTTP** runs over **TCP port 80** and is a **stateless** request/response protocol.
- **HTTPS** runs over **TCP port 443** using **TLS**.
- There are **5+** HTTP methods: **GET**, **POST**, **PUT**, **DELETE**, and **HEAD**.
- **GET** is **idempotent/safe**; **POST** is **not** idempotent.
- HTTP status codes have **5** classes: **1xx**, **2xx**, **3xx**, **4xx**, **5xx**.
- Status **200** means **OK** and **404** means **not found**.
- Status **301** is **permanent redirect** and **304** is **not modified**.
- Status **500** indicates a **server error** and **4xx** a **client error**.
- **HTTP/1.0** uses **non-persistent** connections (**1** per object).
- **HTTP/1.1** uses **persistent** connections by default with **pipelining**.
- HTTP state is added by **cookies** (client-side) and **sessions** (server-side).

File Transfer (FTP)

- **FTP** uses **2** connections: **control port 21** and **data port 20**.
- FTP has **2** modes: **active** (server connects back) and **passive** (client connects).
- **Passive** FTP mode is **firewall/NAT**-friendly.
- FTP supports **2** transfer types: **ASCII** (text) and **binary** (raw).
- FTP control connection is **out-of-band** and stays **open** for the session.

Other Services

- **Telnet** provides **plaintext** remote login on **TCP port 23**.
- **SSH** provides **encrypted** remote login on **TCP port 22**, replacing Telnet.
- **SNMP** manages networks using **UDP ports 161** (agent) and **162** (trap).
- A **proxy** caches responses to reduce **latency** and **bandwidth**.
- There are **2** proxy types: **forward** (client side) and **reverse** (server side).

Network Security Foundations

- The **CIA** triad consists of **3** goals: **confidentiality**, **integrity**, and **availability**.
- The full security goal set adds **2** more: **authentication** and **non-repudiation**.
- **Confidentiality** is provided by **encryption**.
- **Integrity** is provided by **hashing** or **MAC**.
- **Non-repudiation** is provided by **digital signatures**.
- There are **2** attack categories: **passive** (eavesdropping) and **active** (modification).
- **Passive** attacks include **traffic analysis** and are hard to **detect**.
- **Active** attacks include **masquerade**, **replay**, **modification**, and **DoS**.

Cryptography

- There are **2** cryptography types: **symmetric** (secret-key) and **asymmetric** (public-key).
- **Symmetric** cryptography uses the **same** key and is **fast**.
- **Symmetric** key management for n users needs **n(n-1)/2** keys.
- **Asymmetric** cryptography uses a **key pair** and needs only **2n** keys for n users.
- There are **2** symmetric cipher types: **block** (DES/AES) and **stream** (RC4).
- Common block cipher modes include **ECB**, **CBC**, **CTR**, and **CFB/OFB**.
- **ECB** mode is insecure because identical blocks leak **patterns**.
- **DES** uses a **64**-bit block, **56**-bit key, and **16** Feistel rounds.
- **DES** is now insecure because its **56**-bit key is brute-forceable.
- **3DES** applies DES **3** times for an effective key of **112/168** bits.
- **AES** uses a **128**-bit block and keys of **128/192/256** bits.
- **AES** uses **10/12/14** rounds for **128/192/256**-bit keys respectively.
- **AES** is a **substitution-permutation network**, not a Feistel cipher.
- AES has **4** round operations: **SubBytes**, **ShiftRows**, **MixColumns**, and **AddRoundKey**.
- For **confidentiality**, encrypt with the **receiver's public** key.
- For **authentication**, sign with the **sender's private** key.
- **RSA** security relies on the difficulty of **factoring large primes**.
- In RSA, **n = p x q** and **phi(n) = (p-1)(q-1)**.
- RSA encryption is **C = M^e mod n** and decryption is **M = C^d mod n**.
- The RSA public key is **(e, n)** and the private key is **(d, n)**.
- **Diffie-Hellman** establishes a **shared secret** based on the **discrete logarithm** problem.
- **Diffie-Hellman** is vulnerable to **man-in-the-middle** without authentication.
- **3** common asymmetric algorithms are **RSA**, **Diffie-Hellman**, and **ECC**.

Hashing and Integrity

- A **hash function** produces a **fixed-size** digest and is **one-way**.
- Hash properties include **pre-image resistance**, **collision resistance**, and the **avalanche effect**.
- **MD5** produces a **128**-bit digest and is now **broken**.
- **SHA-1** produces a **160**-bit digest and is **deprecated**.
- **SHA-256** produces a **256**-bit digest and is current.
- A **MAC** uses a **shared secret key** to provide **integrity + authentication**.
- A **MAC** does **not** provide **non-repudiation** because the key is shared.
- **Hashing** alone provides only **integrity**, not confidentiality.

Digital Signatures and Certificates

- A **digital signature** is created by **hashing** then encrypting with the **sender's private** key.
- A digital signature provides **3** services: **authentication**, **integrity**, and **non-repudiation**.
- A digital signature does **not** provide **confidentiality** by itself.
- **3** signature algorithms are **RSA**, **DSA**, and **ECDSA**.
- A **digital certificate** binds a **public key** to an **identity**, signed by a **CA**.
- The certificate standard is **X.509**.
- **PKI** includes **CAs**, **registration authorities**, and a **CRL** for revocation.

SSL/TLS

- **SSL/TLS** sits between the **Application** and **Transport** layers.
- **TLS** provides **3** services: **confidentiality**, **authentication**, and **integrity**.
- **TLS** uses **asymmetric** crypto for the **handshake** and **symmetric** for **bulk data**.
- TLS has **4** sub-protocols: **Handshake**, **Record**, **Alert**, and **Change Cipher Spec**.
- **HTTPS** is HTTP over TLS on **port 443**.

Firewalls and IDS

- There are **3** firewall types: **packet-filtering**, **stateful**, and **application/proxy**.
- A **packet-filtering** firewall is **stateless** and inspects **L3/L4** headers.
- A **stateful** firewall tracks **connection state**.
- An **application/proxy** firewall inspects **L7** content.
- A **DMZ** is a buffer subnet for **public-facing** servers.
- **IDS** detects and **alerts** (passive); **IPS** detects and **blocks** (inline).
- There are **2** detection methods: **signature-based** (known) and **anomaly-based** (unknown).
- **Anomaly-based** detection catches **zero-day** attacks but has more **false positives**.

Common Attacks

- A **DoS/DDoS** attack targets **availability** by flooding resources.
- A **MITM** attack intercepts traffic and is countered by **TLS**.
- A **replay** attack resends valid messages and is countered by a **nonce/timestamp**.
- **ARP poisoning** sends fake ARP replies and works only on the **local LAN**.
- **DNS spoofing** injects false **DNS records** into a cache.
- A **SYN flood** exploits the **3-way handshake** with half-open connections, countered by **SYN cookies**.
- A **Smurf** attack uses **ICMP** broadcast amplification.
- **SQL injection** and **XSS** are **application-layer** attacks.
- **Phishing** tricks users into revealing **credentials**.
- **Brute force** attacks guess **keys** or **passwords** exhaustively.

Port Numbers Rapid Fire

- **FTP** uses ports **20** (data) and **21** (control) over **TCP**.
- **SSH** uses port **22** over **TCP**.
- **Telnet** uses port **23** over **TCP**.
- **SMTP** uses port **25** over **TCP**.
- **DNS** uses port **53** over **UDP and TCP**.
- **DHCP** uses ports **67** (server) and **68** (client) over **UDP**.
- **TFTP** uses port **69** over **UDP**.
- **HTTP** uses port **80** over **TCP**.
- **POP3** uses port **110** over **TCP**.
- **NTP** uses port **123** over **UDP**.
- **IMAP** uses port **143** over **TCP**.
- **SNMP** uses ports **161** and **162** over **UDP**.
- **BGP** uses port **179** over **TCP**.
- **HTTPS** uses port **443** over **TCP**.

- **RIP** uses port **520** over **UDP**.
- **IMAPS** uses port **993** and **POP3S** uses port **995**.
- The **IP Protocol** field uses **1** for ICMP, **6** for TCP, **17** for UDP, and **89** for OSPF.

Formula Rapid Fire

- **Transmission delay** equals **L/B** (packet size over bandwidth).
- **Propagation delay** equals **d/v** (distance over speed).
- The ratio **a** equals **Tp/Tt**.
- **Stop-and-Wait efficiency** equals **1/(1+2a)**.
- **Sliding window efficiency** equals **N/(1+2a)**.
- **Optimal window size** equals **1 + 2a**.
- **Bandwidth-delay product** equals **Bandwidth x RTT**.
- **Nyquist bit rate** equals **2B log2(L)**.
- **Shannon capacity** equals **B log2(1 + S/N)**.
- **Bit rate** equals **baud rate x log2(L)**.
- **Usable hosts** per subnet equals **2^h - 2**.
- **Number of subnets** equals **2^s**.
- **Subnet block size** equals **256 - mask octet**.
- **Mesh links** equal **n(n-1)/2**.
- **Go-Back-N window** equals **2^m - 1**.
- **Selective Repeat window** equals **2^(m-1)**.
- **Pure ALOHA** max efficiency equals **18.4%** (**1/2e**).
- **Slotted ALOHA** max efficiency equals **36.8%** (**1/e**).
- **Symmetric keys** for n users equal **n(n-1)/2**.
- **Asymmetric keys** for n users equal **2n**.
- A code's detection power needs **d_min >= s+1** and correction needs **d_min >= 2t+1**.
- The Hamming redundant-bit rule is **2^r >= m + r + 1**.

Header Sizes Rapid Fire

- The **IPv4** header is **20-60** bytes.
- The **TCP** header is **20-60** bytes.
- The **UDP** header is a fixed **8** bytes.
- The **IPv6** header is a fixed **40** bytes.
- The **Ethernet** header is **14** bytes plus a **4**-byte FCS.
- The **ATM cell** is a fixed **53** bytes.
- The MAC address is **48** bits and the IPv4 address is **32** bits.
- The IPv6 address is **128** bits and a port number is **16** bits.

Layer-to-Protocol Rapid Fire

- The **Physical** layer devices include the **hub** and **repeater**.
- The **Data Link** layer devices include the **switch** and **bridge**.
- The **Network** layer device is the **router**, running **IP**, **ICMP**, and **ARP**.
- The **Transport** layer runs **TCP** and **UDP**.

- The **Application** layer runs **HTTP**, **FTP**, **SMTP**, **DNS**, and **DHCP**.
- **ARP** operates between layers **2** and **3**.
- **Routing** occurs at layer **3**; **framing** occurs at layer **2**.
- **Encryption** and **compression** occur at layer **6** (Presentation).
- **Flow control** appears at **2** layers: **Data Link** (node) and **Transport** (end-to-end).

Concept Recap Rapid Fire

- **Switch** learns **source** MAC and forwards by **destination** MAC.
- **Hub** has **1** collision domain; **switch** has **1** per port.
- **Router** separates **broadcast** domains; **switch** does not.
- **TCP** handshake is **3-way**; teardown is **4-way**.
- **Slow start** grows **exponentially**; **congestion avoidance** grows **linearly**.
- **Flow control** protects the **receiver**; **congestion control** protects the **network**.
- **CSMA/CD** is for **wired** Ethernet; **CSMA/CA** is for **wireless** Wi-Fi.
- **Go-Back-N** retransmits **all** after loss; **Selective Repeat** retransmits **only** the lost frame.
- **Distance Vector** uses **Bellman-Ford**; **Link State** uses **Dijkstra**.
- **RIP** max hop count is **15** with **16** meaning unreachable.
- **ARP** maps **IP to MAC**; **RARP** maps **MAC to IP**.
- **DNS** normally uses **UDP** but switches to **TCP** for zone transfers.
- **POP3** deletes mail; **IMAP** keeps and syncs it.
- **Active** FTP has the server connect back; **passive** FTP has the client connect.
- **GET** is idempotent; **POST** is not.
- **Symmetric** crypto is **fast**; **asymmetric** is **slow**.
- For **confidentiality** use the **receiver's public** key; for **signing** use the **sender's private** key.
- **Hashing** gives integrity; **encryption** gives confidentiality; **signatures** give non-repudiation.
- **IDS** alerts; **IPS** blocks.
- **TLS** uses **asymmetric** for handshake and **symmetric** for bulk data.
- A **/30** subnet gives **2** usable hosts for point-to-point links.
- The fragment **offset** is in **8**-byte units.
- **MF=1** means more fragments; **MF=0** marks the **last** fragment.
- **TTL=0** causes the router to drop the packet and send ICMP **Time Exceeded**.
- **IPv6** has **no** header checksum and routers do **not** fragment.
- The Internet checksum uses **1's complement** arithmetic.
- **Manchester** encoding is **self-clocking** at the cost of **2x** bandwidth.
- **Bit stuffing** adds a **0** after **5** ones; the flag is **01111110**.
- **Pure ALOHA** vulnerable time is **2T**; **slotted** is **1T**.
- The minimum Ethernet frame is **64** bytes due to **CSMA/CD**.
- **NAT** with **PAT** maps many hosts to one IP using **ports**.
- **DHCP DORA** = **Discover**, **Offer**, **Request**, **Acknowledge**.
- **Dijkstra** requires **non-negative** edge weights.
- **Circuit switching** reserves bandwidth even when **idle**.
- The **BDP** represents the maximum number of **bits in flight**.
- **Stop-and-Wait** has window size **1**.
- The TCP **TIME-WAIT** state lasts **2*MSL**.
- **DES** key is **56** bits; **AES** block is **128** bits.

- **RSA** relies on **factoring**; **Diffie-Hellman** on **discrete log**.
- **MD5** is **128**-bit; **SHA-1** is **160**-bit; **SHA-256** is **256**-bit.
- A **SYN flood** is mitigated by **SYN cookies**.
- A **replay** attack is mitigated by a **nonce** or **timestamp**.
- The **X.509** standard defines digital **certificates**.
- A **firewall** packet filter inspects **IP**, **port**, and **protocol**.
- **VLAN** tagging uses **802.1Q** with a **12**-bit ID.
- **STP** elects a **root bridge** with the lowest **Bridge ID**.
- The OSI model has **7** layers; TCP/IP has **4**.
- A **gateway** works at all **7** layers as a protocol converter.
- **SSH** (port **22**) is the secure replacement for **Telnet** (port **23**).
- **SNMP** uses **UDP** ports **161** and **162**.
- **HTTP** is **stateless**, so **cookies** maintain state.
- The IPv4 **TTL** field is **8** bits with max value **255**.
- **Fast Ethernet** is **100 Mbps** and **Gigabit** is **1 Gbps**.

Extended Concept Drills

- A **LAN** typically uses **broadcast** channels while a **WAN** uses **point-to-point** switched links.
- The **Internet** is best described as a network of **networks** joined by **routers**.
- A **repeater** regenerates a **signal** and operates at the **physical** layer.
- A **bridge** connects **2** LAN segments and filters by **MAC** address.
- A **collision domain** is broken by a **switch**, **bridge**, or **router**.
- A **broadcast domain** is broken only by a **router** or **VLAN**.
- The **preamble** provides bit **synchronization** for the receiver clock.
- A **jam signal** in CSMA/CD ensures all stations detect the **collision**.
- **Binary exponential backoff** chooses a wait from **0 to 2^k - 1** slots.
- **Token passing** provides **deterministic** (bounded) access delay.
- **FDDI** and **Token Ring** are examples of **token passing** networks.
- The **network ID** identifies the **network**; the **host ID** identifies a **device**.
- A subnet mask separates **network** bits (**1**s) from **host** bits (**0**s).
- The **first** usable host equals network address **+1**.
- The **last** usable host equals broadcast address **-1**.
- **CIDR** reduces routing table size through **route aggregation**.
- **Classful** addressing wastes addresses; **CIDR** is more **efficient**.
- An **autonomous system** is identified by a unique **AS number**.
- **OSPF** divides large networks into **areas** for scalability.
- **RIP** is suitable for **small** networks due to its **15**-hop limit.
- The IPv4 **Identification** field is the **same** for all fragments of a datagram.
- IPv4 **options** can extend the header from **20** up to **60** bytes.
- The **DF** flag set means a packet will be **dropped** rather than fragmented.
- **Path MTU discovery** uses the **DF** flag to find the smallest link MTU.
- The **ICMP** message for an unreachable host is **Destination Unreachable**.
- **Source quench** was an ICMP message for **congestion** control (deprecated).
- **Ping** measures both **reachability** and **round-trip time**.
- **Traceroute** on Windows is the command **tracert**.

- A **DHCP** server offers an IP plus the **mask**, **gateway**, and **DNS**.
- **NAT** helps conserve scarce **IPv4** public addresses.
- The **transport** layer adds **port** numbers for **demultiplexing**.
- A **well-known** port like **80** is reserved for standard **server** services.
- **Ephemeral** ports (**49152-65535**) are assigned to **client** processes.
- **UDP** is preferred for **real-time** traffic where retransmission is useless.
- **TCP** guarantees **in-order**, **reliable**, byte-stream delivery.
- The **PSH** flag tells TCP to **push** data to the application immediately.
- The **RST** flag **resets** (aborts) a TCP connection.
- The **URG** flag marks **urgent** data using the **urgent pointer**.
- **MSS** is the maximum **segment size** negotiated during the handshake.
- A **duplicate ACK** indicates a **missing** segment to the sender.
- **Cumulative ACK** acknowledges all bytes up to the **next expected** byte.
- The TCP **checksum** is **mandatory**, unlike the optional UDP IPv4 checksum.
- **DNS caching** reduces both **latency** and server **load**.
- A **TLD** server handles top-level domains like **.com** and **.org**.
- An **authoritative** server holds the definitive records for a **domain**.
- **SMTP** cannot **pull** mail, which is why **POP3** and **IMAP** exist.
- **MIME** adds the **Content-Type** header to identify the data format.
- An HTTP **cookie** is stored on the **client** to maintain session state.
- **HTTP/2** adds **multiplexing** of streams over a single connection.
- A **persistent** HTTP connection avoids repeated **TCP setup** overhead.
- **FTP** uses a separate **data** connection to keep control **commands** responsive.
- **SSH** can also provide secure **tunneling** and **file transfer (SFTP)**.
- **SNMP** relies on a **MIB** database of managed objects.
- **Symmetric** encryption is suited for **bulk** data due to its **speed**.
- **Asymmetric** encryption solves the **key distribution** problem.
- The **public** key can be shared freely; the **private** key stays **secret**.
- A **block** cipher encrypts fixed-size blocks; a **stream** cipher encrypts **bit-by-bit**.
- **CBC** mode chains blocks using an **IV** to hide patterns.
- **Avalanche effect** means a **1-bit** input change flips about **half** the output bits.
- A **collision** occurs when **2** different inputs produce the **same** hash.
- **HMAC** combines a **hash** with a **secret key** for authentication.
- **Non-repudiation** requires the **private** key only the **sender** possesses.
- A **Certificate Authority** signs certificates to vouch for **public keys**.
- A **CRL** lists **revoked** certificates that are no longer trusted.
- **TLS** succeeded **SSL**, which is now considered **insecure**.
- A **stateful** firewall remembers the **connection** context of packets.
- A **proxy** firewall can inspect and filter **application-layer** content.
- **Signature-based** IDS fails against **unknown** (zero-day) attacks.
- A **DDoS** uses **many** compromised machines (a **botnet**) to flood a target.
- **IP spoofing** forges the **source** address to hide the attacker.
- **ARP poisoning** enables a **MITM** attack on a local LAN.
- **Phishing** is a form of **social engineering** attack.
- A **honeypot** is a decoy system used to **detect** or study attackers.
- **Defense in depth** layers **multiple** security controls.

- A **VPN** creates an encrypted **tunnel** over a public network.
- **IPsec** secures IP traffic with **AH** (integrity) and **ESP** (encryption).
- **Kerberos** is an authentication protocol using **tickets** and a **KDC**.
- The **principle of least privilege** grants the **minimum** access needed.
- A **salt** is random data added before hashing to defeat **rainbow tables**.
- **Two-factor authentication** combines **2** factors like **password** and **OTP**.
- **WEP** is an insecure Wi-Fi protocol replaced by **WPA2** and **WPA3**.
- **WPA2** uses **AES**-based **CCMP** encryption.
- The **frame check sequence** detects errors using **CRC**.
- **Latency** is the **delay**; **throughput** is the **rate**.
- **Jitter** is the **variation** in packet delay.
- A **baud** is one **signal change** per second.
- **Attenuation** is the **loss** of signal strength over distance.
- **Crosstalk** is unwanted signal coupling between adjacent **wires**.
- A **guard band** separates frequency channels in **FDM**.
- **Beamforming** focuses a wireless signal toward a specific **receiver**.
- The **hidden terminal** problem motivates **CSMA/CA** and **RTS/CTS**.
- **RTS/CTS** reserves the wireless medium before **data** transmission.
- **Half-open** TCP connections accumulate during a **SYN flood**.
- **Sequence numbers** prevent acceptance of **duplicate** frames.
- A **checksum** of **all 1s** at the receiver means **no error** detected.
- The **Data Offset** (HLEN) field gives the TCP header length in **4-byte** words.
- **IGMP** (protocol **2**) manages **multicast** group membership.
- **Multicast** sends to a **group**; **broadcast** sends to **all**; **unicast** to **one**.
- **Anycast** delivers to the **nearest** of a group of receivers.
- The **loopback** interface lets a host communicate with **itself**.
- An **IPv4** address with all host bits **0** cannot be assigned to a **host**.
- **Subnetting** a **/24** into **/26**s creates **4** subnets of **62** hosts each.
- A **/26** mask is **255.255.255.192** with block size **64**.
- A **/27** mask is **255.255.255.224** with block size **32**.
- A **/28** mask is **255.255.255.240** with **14** usable hosts.
- **VLSM** requires a **classless** routing protocol like **OSPF**.